

This note provides additional information on the parameter sets of **SQIsignTriangle** with respect to the technical requirements of the algorithm call.

The submitted **SQIsignTriangle** algorithm specifies four parameter sets, together with their corresponding classical and quantum security levels, in **Table 6: SQIsignTriangle Parameter Sets**. The first parameter set provides **128-bit classical security and 64-bit quantum security**. This parameter set is included primarily for **prototype implementation, functional validation, and performance evaluation**, providing a relatively small-scale parameter setting for implementation purposes. It is **not designated as a recommended parameter set corresponding to the security levels required by the algorithm call**.

According to the technical requirements of the algorithm call, an algorithm shall support at least three classical security levels of **128 bits, 256 bits, and 512 bits**, with the corresponding quantum security levels being no less than **80 bits, 128 bits, and 256 bits**, respectively. The **second, third, and fourth parameter sets** in Table 6 provide classical/quantum security levels of **160/80 bits, 256/128 bits, and 512/256 bits**, respectively. In particular, the second parameter set provides **160-bit classical security**, which is higher than the required 128-bit classical security level, together with the required **80-bit quantum security**. The third and fourth parameter sets provide **256/128-bit** and **512/256-bit** classical/quantum security, respectively. Therefore, these three parameter sets correspond to and satisfy the three security levels specified in the technical requirements of the algorithm call.

To distinguish the intended purposes of the parameter sets more clearly, the **128/64-bit** parameter set should be regarded as an **Implementation/Performance Evaluation Parameter Set**, while the **160/80-bit, 256/128-bit, and 512/256-bit** parameter sets constitute the **Recommended Parameter Sets** corresponding to the three required security levels.

Accordingly, the recommended parameter sets of **SQIsignTriangle** satisfy the classical and quantum security requirements specified in the technical requirements of the algorithm call.